

Pixel Based Brain Tumor Detection

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Abstract—Pixel Based Machine Learning technique manages each pixel as a data point from a Magnetic Resonance Image (MRI) scan for analysis and classification. Considering the Brain Tumor detection application, it is important that we identify details and analyze each pixel in isolation to pinpoint every minute and irregularly shaped affected region of the brain.

As a result, to build such a machine learning model, in reality we have a need for significant amount of data of multiple subjects, ensuring precision and generalization. However, Pixel Based Brain Tumor Detection with MRI data of multiple subjects can be a root of significant discrepancies in special context and is susceptible to noise due to the heterogeneity of brain structures and tumor characteristics. And hence to handle such complications we need Machine Learning Technique that is trained in such an environment.

In this paper, we utilized the image filtering-based feature extraction techniques using 10 image filters alongside a pixel-based brain tumor detection technique where the model is trained using Random Forest classifier on multiple subjects so that it can predict tumor independent to brain structure.

Index Terms—image processing, thresholding, feature space, label, magnetic resonance imaging (MRI), Brain Tumor Segmentation dataset (BraTS)

I. INTRODUCTION

Magnetic resonance imaging has become an essential resource for medical imaging in detecting, diagnosing and treating various conditions, providing detailed diseases and anatomical structures.

However, the reliability of MRI-based brain tumor detection remains a critical challenge. Due to misleading of complex and changing appearance of different brain structures. Thus, we focused on addressing this challenge, exploring different medical image analysis techniques with a significance on the blend of feature space creation with various, image processing techniques (filters) and Pixel based detection technique.

The previous study “Pixel Level Feature Space Modeling and Brain Tumor Detection using Machine Learning” [3], utilizes image filtering-based feature extraction techniques using 8 image filters applying a pixel-based brain tumor detection technique on a single volumetric dataset of Brain Tumor Segmentation (BraTS) 2015 dataset. Which was changed to BraTS2020 dataset because, BraTS data since BraTS’17 being significantly different for better from the previous BraTS data (i.e., 2016 and backwards). The data used during BraTS’14-’16 (from TCIA) is said to be discarded,

describing it a mixture of pre- and post-operative scans, and their ground truth labels have been annotated by the fusion of segmentation results from algorithms that ranked highly during BraTS’12 and’13 [4]. In BraTS 2020 dataset we are specifically leveraging the T2 Fluid Attenuated Inversion Recovery (T2-FLAIR) MRI modality and the segmentation data, this research draws from a larger and more diverse patient dataset to improve the model’s generalization and robustness against local heterogeneity brain structure and tumor characteristics.

With the help of image processing filters, images properties are modified by altering the values of pixels based on the pixels next to them, these techniques include applying various filters, noise reduction, edge detection, feature extraction, and segmentation. Preparing the images for feature space creation, better training and further analysis. These feature spaces are multi-dimensional and in-depth representations of the characteristics of these images, which makes it easier to create pixel-based brain tumor detection models that are more reliable and accurate. Pixel-based brain tumor detection, an analysis of MRI images at a granular level to differentiate between tumor and non-tumor regions, for which machine learning models are trained to classify each pixel based on its pixel values that indicate the presence of a tumor. Analyzing each pixel ensures that no details are missed for more accurate tumor detection. In the study I chose to use random forest, a machine learning model that uses a collection of decision trees to make predictions.

In medical imaging the accuracy of detecting tumor and non-tumor regions cannot be overstated. The all-embracing objective is to enhance the depiction of tumor regions within magnetic resonance imaging (MRI) scans, ultimately bridging gaps in current medical imaging practices.

Precise identification of these tumor regions is critical for timely and accurate medical interventions. With the combination of diverse methodologies ranging from pixel-based detection technique to advanced image filters and applying it to a robust dataset, we achieve a comprehensive comparison of the drawbacks and possibilities in medical image analysis. This study aims to contribute valuable insights to the field of medical image processing and

promote improvements in the MRI's capability to detect and characterize tumors.

II. DATASET

This study leverages Brain Tumor Segmentation dataset (BraTS) 2020 obtained from Kaggle as it is a newer version and has a larger dataset with more patient cases and diversity of models trained on it. [1]

The data initially was obtained in NIfTI (.nii or .nii.gz) format zipped files and was extracted. Neuroimaging Informatics Technology Initiative (NIfTI file format) is volumetric data that provides a flexible, broad-based and common file format used in neuroimaging data to store and exchange medical image data, especially in the field of magnetic resonance imaging (MRI). For this study the first 20 volumetric data (i.e., 20 different patient cases) have been considered. Leveraging the T2-Fluid Attention Recovery (FLAIR) and segmented images containing the Ground truth.

A. Data description

The dataset is divided into two files, MICCAI_BraTS2020_TrainingData for training dataset and MICCAI_BraTS2020_ValidationData for validation dataset. The Training file contains 369 folders, each having four multi-modal MRI scans and a segmentation file with ground truth available in NIfTI format. The Validation file contains 125 folders with each having four multi-modal MRI scans in NIfTI format. For this experimental study 20 different subject's volumetric data from the training have been considered.

The 4 MRI modalities are:

- T1-weighted (T1),
- T1-weighted Contrast Enhanced (T1wCE)
- T2-weighted (T2w)
- T2-Fluid Attention Recovery (FLAIR).

The benefit of having multiple alternative MRI modalities is its complementary information, since the brain tumor has multiple sub-regions - Whole tumor, Tumor core, Enhancing tumor. T2weighted (T2w) and T2-Fluid Attention Recovery (FLAIR) are suitable to detect the tumor with the pretemporal edema. Whereas T1-weighted (T1) and T1weighted Contrast Enhanced (T1wCE) can detect the tumor core without pretemporal edema.

The segmentation files contain Ground truth which is meticulously annotated and verified content about the structures or features within an image. Which is usually determined via., manual labeling by a professional. Offering a precise identification of tumor and non-tumor regions.

The Ground truth (GT) provided in the BraTS dataset has 5 lable:

- Label 0 :- Unlabeled volume

- Label 1 :- Necrotic and non - enhancing tumor core (NCR/NET)
- Label 2 :- Peritumoral edema (ED)
- Label 3 :- Missing
- Label 4 :- Enhancing tumor (ET)

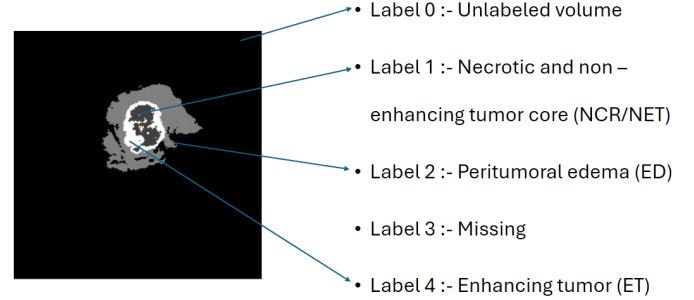


Fig. 1. Ground truth label Naming

According to a statement in the paper [2] It is well known that in the given dataset label 3 is missing. And to emphasize which label corresponds to which label we are representing a figure that highlights the labels.”



Fig. 2. Ground truth labels

B. Data Loading

Deciding on an efficient loading procedure was curucial to manage the volumetric MRI data. NiBabel library is commonly used to perform read and write functions on neuroimaging files formats like NIfTI. Where the image data is accessed via Numpy array. Using libraries like NumPy and OpenCV made the data loading process faster, providing easy access. These tools facilitated easy access to individual frames within the MRI scans, providing for subsequent processing and analysis. The loading process began with 3D data from the training dataset using NiBabel and converted it into 2D data with the help of NumPy and visualized it using OpenCV.

Two different methods were used to save all the frames of Flair and Segmentation Images from the BraTS 3D data verifying the correctness Iterated through each frame in 3D image of flair and segmentation data, save the images with the help of write operation from OpenCV library, giving a file path to systematically storing the output images in specified directories was the first method that was followed. And as the Second method two loops were used. The first loop generated

CSV files from the respective frames and organized them into specified directories. The second loop reads these CSV files, converting them into NumPy arrays, and then saves them as PNG images. (fig.3)

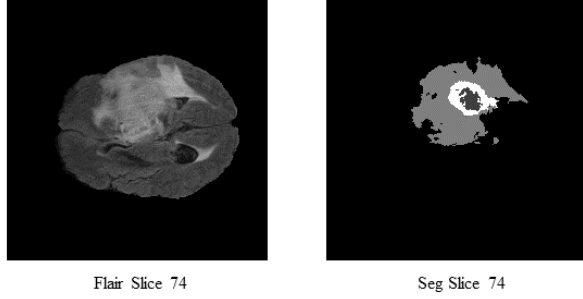


Fig. 3. 74th frame of Flair and Seg Images from Volumetric data 001

C. Data Preprocessing

The volumetric data was carefully prepared to ensure suitability for tumor detection. Which included data cleaning, correcting any unusual patterns errors present in the magnetic resonance imaging (MRI) scans.

Normalization techniques to enhance the constancy of the data by standardizing the intensity levels of the images. Despite random forest, a tree-based model is insensitive to the scale of features and does not necessarily require normalization. It was important for a faster and better performing model as we have a large range of values across the features.

The resolution was also standardized to 8 bits to make the spatial chromaticistics of the MRI scans consistent, limiting the dynamic range of pixel value between 0 and 255. Moreover, granting faster processing of images and optimization of storage space.

Additionally, the images were converted to gray scale using OpenCV, where the pixels are represented by a single intensity value improving the quality of the dataset overall and making future analysis easier.

D. 4th label extraction

The fourth label, enhancing tumor (ET) representing tumor regions is extracted through binary masking approach, a segmentation technique which isolates the region of interest highlighting the enhancing tumor (ET) providing a clear portrayal of tumor boundaries.

The enhancing tumor images are generated by iterating through frames that contain the tumor region of all segmentation volumetric data. For each frame of all 20 volumetric data, regions labeled 4 (enhancing tumor (ET)) is isolated which creates a binary mask ('label'). The resulting images highlight areas labeled 4 in the segmentation data. (fig.4)



Fig. 4. 74th frame of 4th Label Image from volumetric data 001

III. METHODOLOGY

Methodology section explains the outline of the key steps of our project, form data acquisition, preprocessing, Image processing techniques, Balancing dataset, evaluation, model development and extension to volumetric data. We have illustrated our approach in Fig. 5.

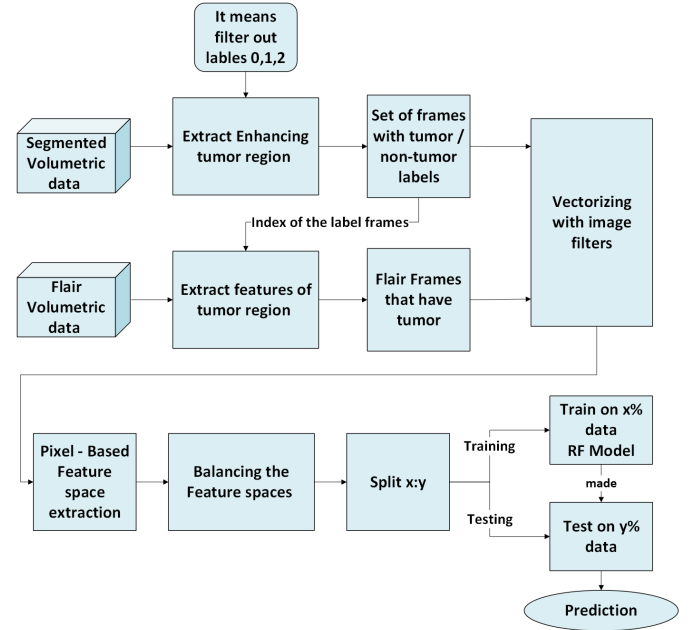


Fig. 5. Methodology Flowchart

A. Image Processing Filters

Initially, we started by applying 8 filters to the flair images in order to extract specific features from images, such as edges, gradients, and textures (fig.6). The eight filters were Gaussian blurring, Canny edge detection, Sobel x and y operators, Laplacian filtering, Histogram of Oriented Gradients (HOG), gradient magnitude and gradient direction calculations. Eventually we chose six more filters out of

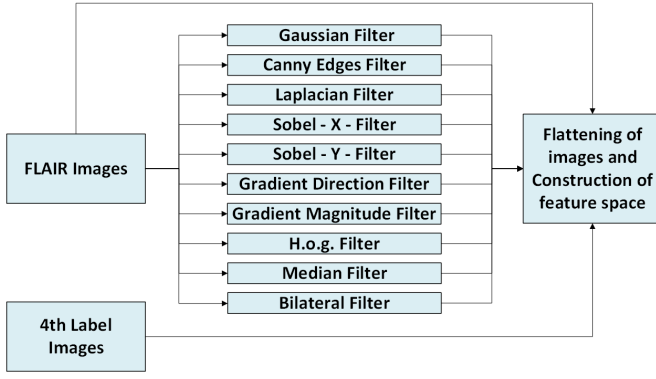


Fig. 6. Image Processing Technique Flowchart

which we selected 2 with the help of feature importance matrices, which are Median filter and Bilateral filter, to enhance different aspects of the image, such as noise reduction, edge preservation, and texture enhancement. The additional two filters expanded the range of features extracted from the images, to potentially improve the performance of the tumor detection algorithm by providing more robust and discriminative features for classification. (fig.7)

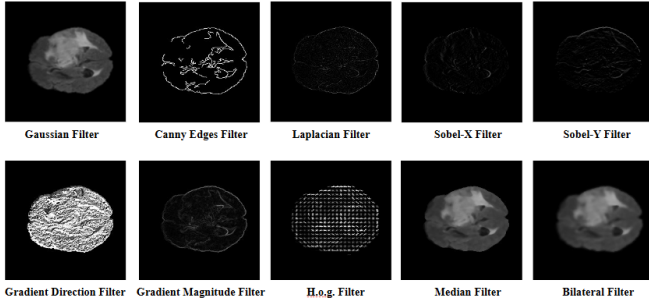


Fig. 7. Image Processing Filters

B. Construction of feature space

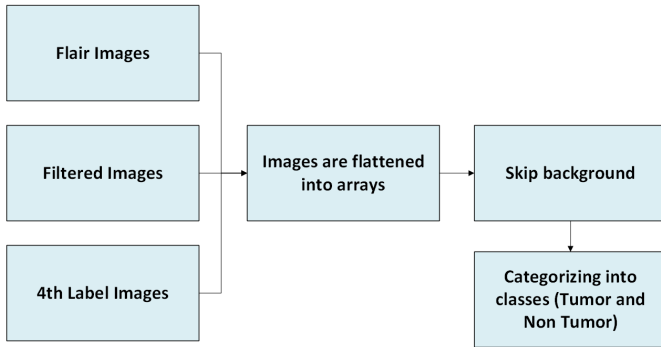


Fig. 8. Feature space construction Flowchart

To represent the extracted feature in a systematic way, we construct the feature space (fig.8). Once the features are

extracted, they are flattened and represented as arrays, where each element corresponds to a specific feature. Each data point in the feature space corresponds to a region in the MRI scan, and the position of the point is determined by the values of the features extracted from that region.

The feature space construction started with ensuring that the flair images, filtered images and the fourth label images all have the same dimensions. For construction of feature spaces, we then iterated through each pixel of the flair, filtered and fourth label images, categorizing pixels based on their values. Background pixels (where both flair and label values are 0) were skipped. Pixels with nonzero value in flair image and zero values in label image were labeled as "no tumor". While pixels with both nonzero values in both flair and label image were labeled as "tumor". This feature space was constructed by storing the nonzero flair pixel values as f1 along with their corresponding labels and the features from f2 to f9 or f11 for 8 or 10 features respectively. The feature space was organized into a structured dataframe for further analysis. The resulting dataframe is saved as a CSV file. This process was followed on a loop to build a feature space for all frames across 20 volumetric data.

C. Balancing Dataset

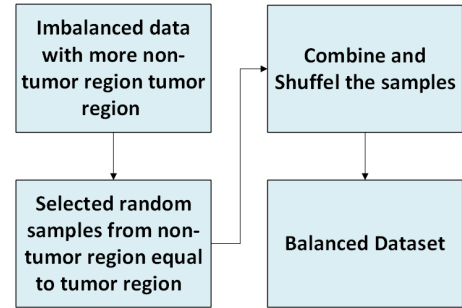


Fig. 9. Balancing dataset Flowchart

To make sure that we have an equal representation of both tumor and non-tumor samples we balanced the dataset(fig.9), as an imbalance can lead to biased models that perform poorly on the minority class (tumor) while achieving high accuracy on the majority class (non-tumor). The initial feature spaces that we generated and used for training the models were imbalanced, having fewer tumor samples compared to non-tumor samples. This class imbalance led to the model having high accuracy but poor Precision and Recall for tumor detection. This meant that the model accurately classified non-tumor samples, but found it hard to correctly identify tumor samples, which is critical for medical diagnoses. To address this issue, we balanced the feature spaces through random sampling. Which involved randomly under sampling the majority class (non-tumor) to match the number of samples in the minority class (tumor). Created a more

balanced dataset, which in turn ensures that the models learn from both tumor and non-tumor samples equally, improving their ability to generalize to new, unseen data. This led to more accurate and reliable predictions for tumor detection.

D. Random Forest

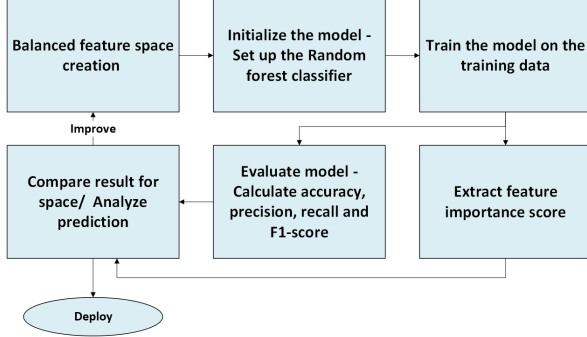


Fig. 10. Random Forest Flowchart

The segmentation images of all 20 volumetric data from the BraTS 2020 dataset were first processed, and the enhancing tumor region was extracted by filtering out labels 0,1,2, leaving only the set of frames with tumor and non-tumor labels. And from the indices of the 4th label tumor frames, features are extracted from the corresponding FLAIR images. Then the Flair tumor frames are initially vectorized with 8 image filters to extract features and are added later with the help of feature importance matrices to improve the feature extraction results. The pixel-based feature space was generated for each frame across all 20 volumetric datasets and was concatenated to create a comprehensive feature space for each volumetric data. To solve the class imbalance in the feature spaces, a balanced feature space was created by random sampling the majority class (nontumor) to match the minority class (tumor), ensures that the models learn equally from both classes, improving the model's ability to detect tumors generalized on the balanced feature space. A Random Forest classifier was trained on 80% of the first volumetric data that was split and tested on the remaining 20%, as well as across the feature spaces of the other datasets (Fig.5).

IV. EXPERIMENTAL AND RESULT SECTION

(In this study many experiments were constructed which are refereed to as simulation 1, 2, and 3 respectively. Where simulation 1 and 2 mainly .. on the impact of the features on the performances and the efficiency. And simulation 3 explores on training the first volumetric data and testing ot on others.)

A. Simulation 1

In the first simulation, we consider twenty feature spaces, each containing 8 features along with a label. Twelve of these data samples were concatenated into a single feature space, which was retained as the training data. This training data

was then used to train a Random Forest classifier, having a training F1-score of 99.73% (TABLE I). Eventually the trained model was tested on the remaining 8 volumetric data samples (volumetric data 013 to 020), yielding F1-score of 92.09%, 78.22%, 88.01%, 80.10%, 91.52%, 86.91%, 93.96% and 82.82%, respectively, with an average F1-score of 86.70% (TABLE II).

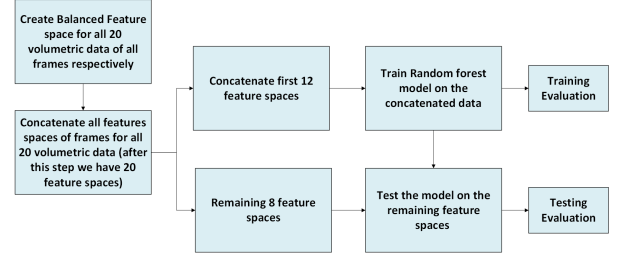


Fig. 11. Simulation 1 and 2 Flowchart

TABLE I
TRAINING EVALUATION OF FIRST 12 VOLUMETRIC DATA CONCATENATED (8 FEATURES)

Training Evaluation	Accuracy	Precision	Recall	F1-Score
First 12 VD	99.73%	99.73%	99.73%	99.73%

TABLE II
TESTING EVALUATION OF THE REMAINING 8 VOLUMETRIC DATA (8 FEATURES)

Testing Evaluation	Accuracy	Precision	Recall	F1-Score
VD013	92.10%	92.40%	92.10%	92.02%
VD014	78.82%	82.34%	78.82%	78.22%
VD015	88.08%	89.01%	88.08%	88.01%
VD016	80.33%	81.83%	80.33%	80.10%
VD017	91.52%	92.31%	91.52%	91.52%
VD018	87.08%	89.04%	87.08%	86.91%
VD019	93.96%	93.99%	93.96%	93.96%
VD020	82.83%	82.89%	82.83%	82.82%

B. Simulation 2

To study the impact of varying number of features, we conducted a second simulation, which included an additional 2 features based on the feature importance matrices (Bilateral filter and Median filter). In this simulation we again considered the twenty volumetric data but with 10 features and the label. Following the same process, twelve feature spaces were concatenated to make training data, which was again used to train the Random Forest classifier, having a training F1-score of 99.95% (TABLE III). The trained model was then tested on the remaining 8 feature space yielding us a F1-score of 93.69%, 78.44%, 89.14%, 80.22%, 92.16%, 86.88%, 94.97% and 83.74% respectively with an average F1-score of 87.53% (TABLE IV).

Demonstrating the addition of 2 relevant features in the second simulation led to a modest improvement in both training

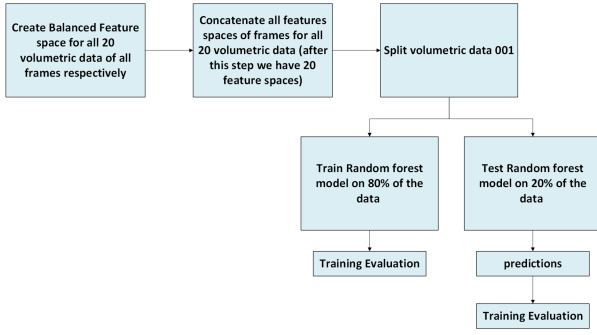


Fig. 12. Simulation 3 Flowchart

TABLE III

TRAINING EVALUATION OF FIRST 12 VOLUMETRIC DATA CONCATENATED (10 FEATURES)

Training Evaluation	Accuracy	Precision	Recall	F1-Score
First 12 VD	99.95%	99.95%	99.95%	99.95%

TABLE IV

TESTING EVALUATION OF THE REMAINING 8 VOLUMETRIC DATA (10 FEATURES)

Testing Evaluation	Accuracy	Precision	Recall	F1-Score
VD013	93.70%	93.99%	93.70%	93.69%
VD014	79.06%	82.77%	79.06%	78.44%
VD015	89.21%	90.23%	89.21%	89.14%
VD016	80.38%	81.38%	80.38%	80.22%
VD017	92.20%	93.10%	92.20%	92.19%
VD018	87.05%	89.07%	87.05%	86.88%
VD019	94.97%	95.03%	94.97%	94.97%
VD020	83.75%	83.82%	83.75%	83.74%

and testing F1-scores, indicating the model's performance on unseen data and improved generalization.

In the third simulation to further explore the model performances, we conducted two experiments using 20 volumetric data like the previous 2 simulations.

C. Simulation 3

(curiosity in 2 para.,s)

In the third simulation out of curiosity we conducted 2 more experiments to further analyse the importance of the features

In the first experiment, volumetric data 001 with 8 features was taken a split in an 80:20 ratio, 80% is used as the training data and was trained with the Random Forest classifier achieving a training F1-score of 99.94% (TABLE V). The model was tested on the remaining 20% data, giving an F1-score of 90%. Additionally, to assess the generalization of the model, it was also tested on the other 19 volumetric data. The results have varied performances, with F1-scores of 78%, 28%, 86%, 70%, 86%, 83%, 84%, 57%, 48%, 89%, 92%, 85%, 81%, 84%, 76%, 94%, 81%, 90%, 76% (TABLE VI).

In the next experiment, volumetric data 001 was taken but with 10 features. Following the same process the feature space was again split into an 80:20 ratio, and the Random

TABLE V
TRAINING AND TESTING EVALUATION OF 80:20 RATIO OF VOLUMETRIC DATA 001 (8 FEATURES)

	Confusion Matrices	Accuracy	Precision	Recall	F1-Score
Training Evaluation	[[4779 777] [305 5236]]	99.94%	99.94%	99.94%	99.94%
Testing Evaluation		99.95%	99.95%	99.95%	99.95%

TABLE VI

TESTING EVALUATION OF THE REMAINING 19 VOLUMETRIC DATA (8 FEATURES)

Testing Evaluation	Accuracy	Precision	Recall	F1-Score
VD002	78.32%	79.32%	78.32%	78.14%
VD003	28.40%	27.82%	28.40%	27.93%
VD004	85.99%	86.96%	85.99%	85.90%
VD005	72.21%	81.47%	72.21%	70.00%
VD006	85.99%	86.66%	85.99%	85.93%
VD007	82.94%	84.74%	82.94%	82.71%
VD008	83.98%	86.93%	83.98%	83.65%
VD009	61.73%	71.41%	61.73%	56.85%
VD010	57.12%	76.65%	57.12%	47.50%
VD011	89.44%	90.40%	89.44%	89.37%
VD012	92.24%	92.71%	92.24%	92.22%
VD013	85.24%	86.71%	85.24%	85.09%
VD014	81.18%	81.78%	81.18%	81.09%
VD015	84.05%	84.64%	84.05%	83.98%
VD016	76.21%	77.57%	76.21%	75.92%
VD017	93.62%	93.79%	93.62%	93.61%
VD018	81.21%	82.48%	81.21%	81.02%
VD019	90.53%	91.04%	90.53%	90.50%
VD020	76.82%	78.95%	76.82%	76.38%

Forest Classifier was trained on 80% of the data, achieving a slightly higher training F1-score of 99.96% (TABLE VII). The model was tested on the remaining 20% of the data, yielding an F1-score of 91%. The trained model was also tested on the other 19 volumetric data yielding an F1-score of 71%, 16%, 83%, 68%, 84%, 76%, 86%, 55%, 41%, 89%, 92%, 87%, 79%, 81%, 71%, 94%, 77%, 91%, 74% (TABLE VIII).

TABLE VII

TRAINING AND TESTING EVALUATION OF 80:20 RATIO OF VOLUMETRIC DATA 001 (10 FEATURES)

	Confusion Matrices	Accuracy	Precision	Recall	F1-Score
Training Evaluation	[[4830 726] [233 5308]]	99.96%	99.96%	99.96%	99.96%
Testing Evaluation		91.36%	91.69%	91.36%	91.34%

Here we justify that feature addition may have improved the model's ability to capture relevant patterns in the data based on the Training and testing evaluation. But the model still yielded considerable variation while testing on different

TABLE VIII
TESTING EVALUATION OF THE REMAINING 19 VOLUMETRIC DATA (10
FEATURES)

Testing Evaluation	Accuracy	Precision	Recall	F1-Score
VD002	71.82%	74.82%	71.82%	70.92%
VD003	16.38%	16.23%	16.38%	16.29%
VD004	83.15%	85.08%	83.15%	82.91%
VD005	70.83%	81.03%	70.83%	68.22%
VD006	84.40%	85.51%	84.40%	84.28%
VD007	76.65%	80.74%	76.65%	75.85%
VD008	86.33%	88.76%	86.33%	86.12%
VD009	60.63%	71.50%	60.63%	54.92%
VD010	54.72%	75.76%	53.72%	41.13%
VD011	88.99%	90.15%	88.99%	88.90%
VD012	91.62%	92.28%	91.62%	91.59%
VD013	87.44%	88.58%	87.44%	87.34%
VD014	79.54%	81.04%	79.54%	79.29%
VD015	81.03%	82.33%	81.03%	80.83%
VD016	71.93%	74.46%	71.93%	71.18%
VD017	94.50%	94.73%	94.50%	94.50%
VD018	77.91%	80.48%	77.91%	77.43%
VD019	90.55%	91.09%	91.55%	90.52%
VD020	74.88%	77.78%	74.88%	74.21%

unseen data.

CONCLUSION

When creating simulation 2 randomly using 6 feature spaces the results were poorer compared to simulation 1

simulation 2 results were better when we added 2 top performing features to the original 8 in simulation 1

When comparing simulation 1 and 2 we observe that the results of simulation 2, is slightly better than simulation 1.

The model gives better results if we selectively chose the top performing features.

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